Practical Digital Systems Design

Homework 3:

Logic Synthesis & Analysis of

Combinational Circuits

Name: 尋以諾

Student ID: B123015021

➢ Synthesis Flow

I. Ripple Carry Adder

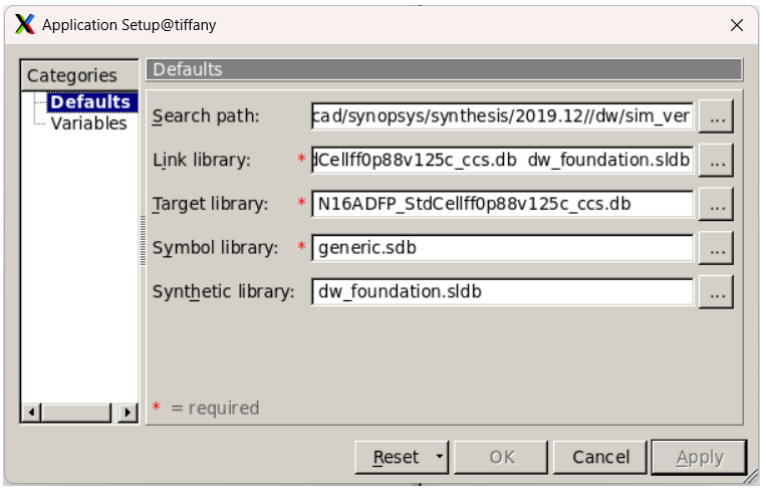
1. Upload the **.synopsys_dc.setup** file to the RCA/ directory via FileZilla.
2. In the Xshell terminal, enter **ls -al** to verify that the correct configuration file is included.

```
[B123015021@tiffany RCA]$ ls -a
.  ..  N16ADFP_StdCell.v  .synopsys_dc.setup  tb.v  top.v
```

3. In the Xshell terminal, enter **ncverilog tb.v +access+r** to compile and simulate top.v, and verify that the simulation runs correctly.



4. In the Xshell terminal, enter **dv &** and wait for Design Vision to launch.
5. In the Toolbox, select **File → Setup** and verify that the library contents are correctly configured.



6. Load the top.v file to be synthesized, and verify that the message **“Presto compilation completed successfully”** is displayed.

```
Compiling source file /home/UGP/B123015021/HW3/HW1/RCA/top.v
Presto compilation completed successfully.
Current design is now '/home/UGP/B123015021/HW3/HW1/RCA/FA.db:FA'
Loaded 2 designs.
Current design is 'FA'.
```

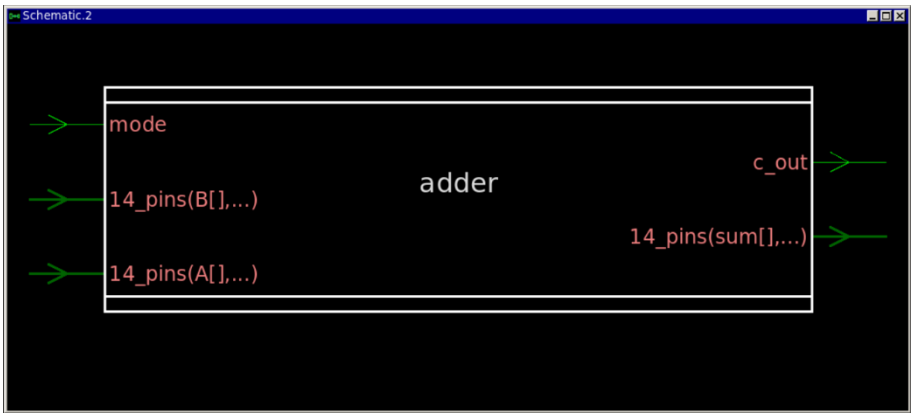
7. Verify that the hierarchy is set to the top module. Otherwise, enter the command **current_design adder** in the command line.

```
design_vision> current_design adder
Current design is 'adder'.
```

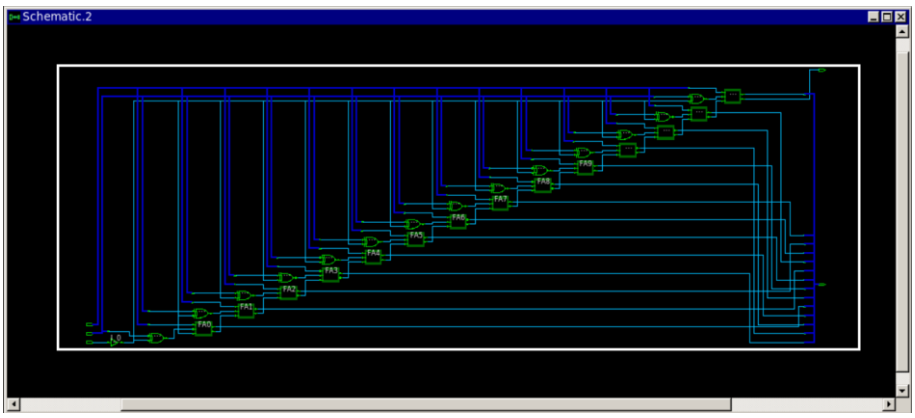
8. In the command line, enter: **set_fix_multiple_port_nets -all -buffer_constants**

```
design_vision> set_fix_multiple_port_nets -all -buffer_constants
1
```

9. View the Symbol View.



10. View the Schematic View.



11. In the Toolbox, select **Design → Compile Design**, and verify that the message **"Optimization Complete"** is displayed.

```
Optimization Complete
---------------------
1
```

12. In the Toolbox, select **Design → Report Area** to view the area report.

```
****************************************
Report : area
Design : adder
Version: Q-2019.12
Date   : Mon Nov 10 19:36:52 2025
****************************************

Library(s) Used:

    N16ADFP_StdCellff0p88v125c_ccs (File: /cad/CBDK/Executable_Packa

Number of ports:                          114
Number of nets:                           156
Number of cells:                           71
Number of combinational cells:             57
Number of sequential cells:                 0
Number of macros/black boxes:               0
Number of buf/inv:                          1
Number of references:                      16

Combinational area:                28.460161
Buf/Inv area:                       0.155520
Noncombinational area:              0.000000
Macro/Black Box area:               0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                   28.460161
Total area:                 undefined

***** End Of Report *****
```

13. In the Toolbox, select **Design → Report Power** to view the power report.

```
  Cell Internal Power  =  23.4616 uW   (73%)
  Net Switching Power  =   8.4660 uW   (27%)
                         ---------
Total Dynamic Power    =  31.9276 uW  (100%)

Cell Leakage Power     =  10.1802 uW

Information: report_power power group summary does not include estimated clock tree power. (PWR-789)

                 Internal         Switching           Leakage            Total
Power Group      Power            Power               Power              Power   (   %    )  Attrs
--------------------------------------------------------------------------------------------------
io_pad             0.0000           0.0000              0.0000             0.0000  (   0.00%)
memory             0.0000           0.0000              0.0000             0.0000  (   0.00%)
black_box          0.0000           0.0000              0.0000             0.0000  (   0.00%)
clock_network      0.0000           0.0000              0.0000             0.0000  (   0.00%)
register           0.0000           0.0000              0.0000             0.0000  (   0.00%)
sequential         0.0000           0.0000              0.0000             0.0000  (   0.00%)
combinational  2.3462e-02       8.4660e-03          1.0180e+04         4.2108e-02  ( 100.00%)
--------------------------------------------------------------------------------------------------
Total          2.3462e-02 mW     8.4660e-03 mW     1.0180e+04 nW     4.2108e-02 mW

***** End Of Report *****
```

14. In the Toolbox, select **Timing → Report Timing Path** to view the delay time from the path's starting point to each node.

    i. max

```
Startpoint: mode (input port)
Endpoint: c_out (output port)
Path Group: (none)
Path Type: max

Des/Clust/Port     Wire Load Model       Library
------------------------------------------------
adder              ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
FA_0               ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
FA_13              ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
FA_12              ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
FA_11              ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
FA_10              ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
FA_9               ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
FA_8               ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
FA_7               ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
FA_6               ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
FA_5               ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
FA_4               ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
FA_3               ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
FA_2               ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
FA_1               ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs

Point                                    Incr       Path
-----------------------------------------------------------
input external delay                     0.00       0.00 r
mode (in)                                0.00       0.00 r
U30/ZN (CKND1BWP16P90)                   0.05       0.05 f
U29/Z (XOR2D1BWP16P90)                   0.02       0.07 r
FA0/B (FA_0)                             0.00       0.07 r
FA0/U3/Z (XOR2D1BWP16P90)                0.02       0.09 f

FA0/U2/Z (AO22D1BWP16P90)                0.02       0.11 f
FA0/c_out (FA_0)                         0.00       0.11 f
FA1/c_in (FA_13)                         0.00       0.11 f
FA1/U2/Z (AO22D1BWP16P90)                0.02       0.14 f
FA1/c_out (FA_13)                        0.00       0.14 f
FA2/c_in (FA_12)                         0.00       0.14 f
FA2/U2/Z (AO22D1BWP16P90)                0.02       0.16 f
FA2/c_out (FA_12)                        0.00       0.16 f
FA3/c_in (FA_11)                         0.00       0.16 f
FA3/U2/Z (AO22D1BWP16P90)                0.02       0.18 f
FA3/c_out (FA_11)                        0.00       0.18 f
FA4/c_in (FA_10)                         0.00       0.18 f
FA4/U2/Z (AO22D1BWP16P90)                0.02       0.20 f
FA4/c_out (FA_10)                        0.00       0.20 f
FA5/c_in (FA_9)                          0.00       0.20 f
FA5/U2/Z (AO22D1BWP16P90)                0.02       0.22 f
FA5/c_out (FA_9)                         0.00       0.22 f
FA6/c_in (FA_8)                          0.00       0.22 f
FA6/U2/Z (AO22D1BWP16P90)                0.02       0.24 f
FA6/c_out (FA_8)                         0.00       0.24 f
FA7/c_in (FA_7)                          0.00       0.24 f
FA7/U2/Z (AO22D1BWP16P90)                0.02       0.26 f
FA7/c_out (FA_7)                         0.00       0.26 f
FA8/c_in (FA_6)                          0.00       0.26 f
FA8/U2/Z (AO22D1BWP16P90)                0.02       0.28 f
FA8/c_out (FA_6)                         0.00       0.28 f
FA9/c_in (FA_5)                          0.00       0.28 f
FA9/U2/Z (AO22D1BWP16P90)                0.02       0.30 f
FA9/c_out (FA_5)                         0.00       0.30 f
FA10/c_in (FA_4)                         0.00       0.30 f


FA10/U2/Z (AO22D1BWP16P90)               0.02       0.33 f
FA10/c_out (FA_4)                        0.00       0.33 f
FA11/c_in (FA_3)                         0.00       0.33 f
FA11/U2/Z (AO22D1BWP16P90)               0.02       0.35 f
FA11/c_out (FA_3)                        0.00       0.35 f
FA12/c_in (FA_2)                         0.00       0.35 f
FA12/U2/Z (AO22D1BWP16P90)               0.02       0.37 f
FA12/c_out (FA_2)                        0.00       0.37 f
FA13/c_in (FA_1)                         0.00       0.37 f
FA13/U2/Z (AO22D1BWP16P90)               0.01       0.38 f
FA13/c_out (FA_1)                        0.00       0.38 f
c_out (out)                              0.00       0.38 f
data arrival time                                   0.38
-----------------------------------------------------------
(Path is unconstrained)
```

ii. min

```
Startpoint: A[13] (input port)
Endpoint: c_out (output port)
Path Group: (none)
Path Type: min

Des/Clust/Port     Wire Load Model       Library
------------------------------------------------
adder              ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
FA_1               ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs

Point                                    Incr       Path
-----------------------------------------------------------
input external delay                     0.00       0.00 f
A[13] (in)                               0.00       0.00 f
FA13/A (FA_1)                            0.00       0.00 f
FA13/U2/Z (AO22D1BWP16P90)               0.01       0.01 f
FA13/c_out (FA_1)                        0.00       0.01 f
c_out (out)                              0.00       0.01 f
data arrival time                                   0.01
-----------------------------------------------------------
(Path is unconstrained)
```

15. In the Toolbox, select **Select → Path From/Through/To**, set the delay type to max, and check the critical path.



16. In the command line, enter: **change_names -hierarchy -rule verilog**

```
design_vision> change_names -hierarchy -rule verilog
1
```

17. In the command line, enter: **report_qor > adder.qor** to save the report.

```
design_vision> report_qor > adder.qor
```

18. In the Toolbox, select **File → Save As**, name the file **adder_syn.v**, and click **Save** to store the synthesized file.
19. In the command line, enter: **write_sdf -version 1.0 -context verilog adder_syn.sdf** to write the timing information into the SDF file.

```
design_vision> write_sdf -version 1.0 -context verilog adder_syn.sdf
Information: Annotated 'cell' delays are assumed to include load delay. (UID-282)
Information: Writing timing information to file '/home/UGP/B123015021/HW3/HW1/RCA/adder_syn.sdf'. (WT-3)
1
```

20. Modify the testbench by commenting out **`include "./top.v"**, then adding **`include "./adder_syn.v"** and **`include "./N16ADFP_StdCell.v"**.

```
1 `timescale 10ns/100ps
2 // `include "./top.v"
3 `include "./adder_syn.v"
4 `include "./N16ADFP_StdCell.v"
```

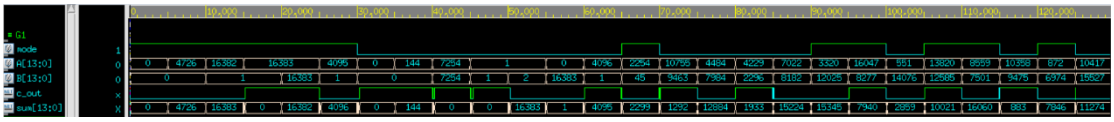
Inside the module, insert **initial $sdf_annotate ("adder_syn.sdf", u_adder);** to apply timing information from the SDF file to the design under test, ensuring that the synthesized design and standard cell library are correctly included for post-synthesis simulation.

```
1 adder u_adder (A, B, mode, sum, c_out);
2 initial $sdf_annotate ("adder_syn.sdf", u_adder);
```

21. In the Xshell terminal, enter **ncverilog tb.v +access+r** to rerun the simulation.



22. In Notepad++, check that the message **"Annotation completed successfully"** is displayed.
23. In the Xshell terminal, enter **nWave &** to launch nWave, open tb, and select the signals you want to observe.



II. Voting Circuit

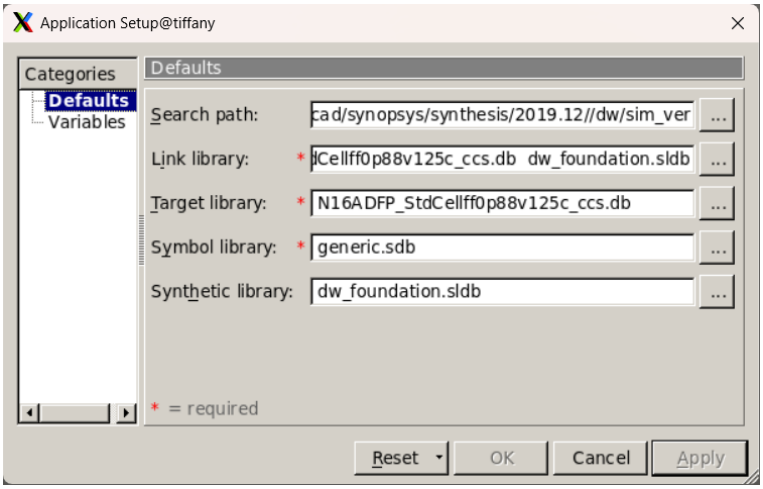
1. Upload the **.synopsys_dc.setup** file to the RCA/ directory via FileZilla.
2. In the Xshell terminal, enter **ls -al** to verify that the correct configuration file is included.

```
[B123015021@tiffany voting]$ ls -a
.  ..  N16ADFP_StdCell.v  .synopsys_dc.setup  tb_voting.v  voting.v
```

3. In the Xshell terminal, enter **ncverilog tb_voting.v +access+r** to compile and simulate top.v, and verify that the simulation runs correctly.



4. In the Xshell terminal, enter **dv &** and wait for Design Vision to launch.
5. In the Toolbox, select **File → Setup** and verify that the library contents are correctly configured.



6. Load the top.v file to be synthesized, and verify that the message **"Presto compilation completed successfully"** is displayed.

```
Compiling source file /home/UGP/B123015021/HW3/HW2/voting/voting.v
Presto compilation completed successfully.
Current design is now '/home/UGP/B123015021/HW3/HW2/voting/ha.db:ha'
Loaded 9 designs.
Current design is 'ha'.
```

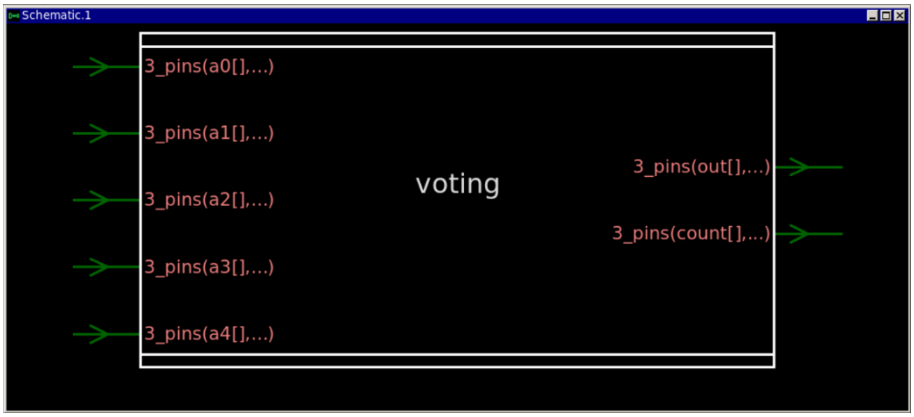
7. Verify that the hierarchy is set to the top module. Otherwise, enter the command **current_design voting** in the command line.

```
design_vision> current_design voting
Current design is 'voting'.
```

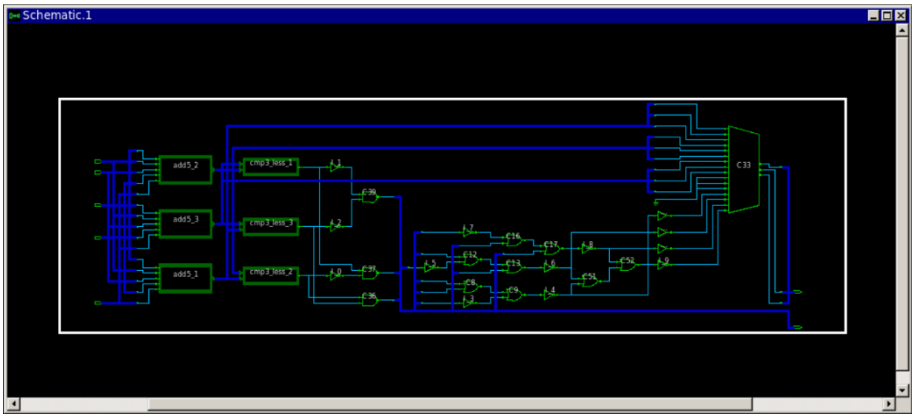
8. In the command line, enter: **set_fix_multiple_port_nets -all -buffer_constants**

```
design_vision> set_fix_multiple_port_nets -all -buffer_constants
1
```

9. View the Symbol View.



10. View the Schematic View.



11. In the Toolbox, select **Design → Compile Design**, and verify that the message **"Optimization Complete"** is displayed.

```
Optimization Complete
---------------------
1
```

12. In the Toolbox, select **Design → Report Area** to view the area report.

```
****************************************
Report : area
Design : voting
Version: Q-2019.12
Date   : Sat Nov 15 01:55:28 2025
****************************************

Information: Updating design information... (UID-85)
Library(s) Used:

    N16ADFP_StdCellff0p88v125c_ccs (File: /cad/CBDK/Executable_Packa

Number of ports:                           66
Number of nets:                           120
Number of cells:                           66
Number of combinational cells:             60
Number of sequential cells:                 0
Number of macros/black boxes:               0
Number of buf/inv:                         12
Number of references:                      11

Combinational area:                18.766081
Buf/Inv area:                       1.866240
Noncombinational area:              0.000000
Macro/Black Box area:               0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                   18.766081
Total area:                 undefined

***** End Of Report *****
```

13. In the Toolbox, select **Design → Report Power** to view the power report.

```
  Cell Internal Power  =  12.0654 uW   (56%)
  Net Switching Power  =   9.3089 uW   (44%)
                         ---------
Total Dynamic Power    =  21.3743 uW  (100%)

Cell Leakage Power     =   5.8527 uW

Information: report_power power group summary does not include estimated clock tree power. (PWR-789)

                 Internal         Switching           Leakage            Total
Power Group      Power            Power               Power              Power   (   %    )  Attrs
--------------------------------------------------------------------------------------------------
io_pad             0.0000           0.0000              0.0000             0.0000  (   0.00%)
memory             0.0000           0.0000              0.0000             0.0000  (   0.00%)
black_box          0.0000           0.0000              0.0000             0.0000  (   0.00%)
clock_network      0.0000           0.0000              0.0000             0.0000  (   0.00%)
register           0.0000           0.0000              0.0000             0.0000  (   0.00%)
sequential         0.0000           0.0000              0.0000             0.0000  (   0.00%)
combinational  1.2065e-02        9.3089e-03          5.8527e+03         2.7227e-02  ( 100.00%)
--------------------------------------------------------------------------------------------------
Total          1.2065e-02 mW     9.3089e-03 mW     5.8527e+03 nW     2.7227e-02 mW
```

14. In the Toolbox, select **Timing → Report Timing Path** to view the delay time from the path's starting point to each node.

    i. max

```
Startpoint: a2[2] (input port)
Endpoint: count[0] (output port)
Path Group: (none)
Path Type: max

Des/Clust/Port     Wire Load Model       Library
------------------------------------------------
voting             ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
add5_1             ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
cmp3_less_1        ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
cmp3_less_0        ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs

Point                                                   Incr       Path
--------------------------------------------------------------------------
input external delay                                    0.00       0.00 f
a2[2] (in)                                              0.00       0.00 f
add5_3/C (add5_1)                                       0.00       0.00 f
add5_3/U10/Z (XOR2D1BWP16P90)                           0.02       0.02 r
add5_3/U9/ZN (ND2D1BWP16P90)                            0.01       0.03 f
add5_3/U8/ZN (OAI21D1BWP16P90)                          0.01       0.04 r
add5_3/U7/Z (XOR2D1BWP16P90)                            0.02       0.06 f
add5_3/U3/ZN (OAI22D1BWP16P90)                          0.01       0.07 r
add5_3/U2/Z (XOR2D1BWP16P90)                            0.03       0.10 f
add5_3/sum[1] (add5_1)                                  0.00       0.10 f
cmp3_less_1/A[1] (cmp3_less_0)                          0.00       0.10 f
cmp3_less_1/U4/ZN (CKND1BWP16P90)                       0.01       0.11 r
cmp3_less_1/U3/Z (AN2D1BWP16P90)                        0.01       0.12 r
```

```
cmp3_less_1/U2/ZN (OAI222D1BWP16P90)                    0.01       0.13 f
cmp3_less_1/U1/ZN (IOA21D1BWP16P90)                     0.01       0.14 r
cmp3_less_1/L (cmp3_less_0)                             0.00       0.14 r
U17/ZN (INR2D1BWP16P90)                                 0.02       0.16 r
U16/ZN (AOI22D1BWP16P90)                                0.01       0.17 f
U15/ZN (IOA21D1BWP16P90)                                0.00       0.17 r
count[0] (out)                                          0.00       0.17 r
data arrival time                                                  0.17
--------------------------------------------------------------------------
(Path is unconstrained)
```

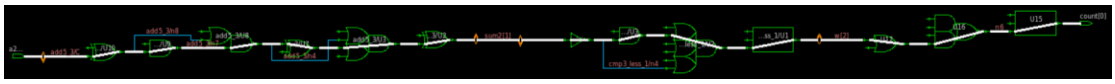
ii. min

```
Startpoint: a3[2] (input port)
Endpoint: count[0] (output port)
Path Group: (none)
Path Type: min

Des/Clust/Port     Wire Load Model       Library
------------------------------------------------
voting             ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
add5_1             ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
cmp3_less_1        ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
cmp3_less_0        ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs

Point                                    Incr       Path
------------------------------------------------------------
input external delay                     0.00       0.00 r
a3[2] (in)                               0.00       0.00 r
add5_3/D (add5_1)                        0.00       0.00 r
add5_3/U11/ZN (CKND1BWP16P90)            0.01       0.01 f
add5_3/U6/Z (XOR2D1BWP16P90)             0.01       0.02 f
add5_3/sum[0] (add5_1)                   0.00       0.02 f
U15/ZN (IOA21D1BWP16P90)                 0.01       0.03 f
count[0] (out)                           0.00       0.03 f
data arrival time                                   0.03
------------------------------------------------------------
(Path is unconstrained)
```

15. In the Toolbox, select **Select → Path From/Through/To**, set the delay type to max, and check the critical path.



16. In the command line, enter: **change_names -hierarchy -rule verilog**

```
design_vision> change_names -hierarchy -rule verilog
1
```

17. In the command line, enter: **report_qor > voting.qor** to save the report.

```
design_vision> report_qor > voting.qor
```

18. In the Toolbox, select **File → Save As**, name the file **voting_syn.v**, and click **Save** to store the synthesized file.

19. In the command line, enter: **write_sdf -version 1.0 -context verilog voting_syn.sdf** to write the timing information into the SDF file.

```
design_vision> write_sdf -version 1.0 -context verilog voting_syn.sdf
Information: Annotated 'cell' delays are assumed to include load delay. (UID-282)
Information: Writing timing information to file '/home/UGP/B123015021/HW3/HW2/voting/voting_syn.sdf'. (WT-3)
1
```

20. Modify the testbench by commenting out **`include "./voting.v"**, then adding **`include "./voting_syn.v"** and **`include "./N16ADFP_StdCell.v"**.

```
1 `timescale 10ns/100ps
2 // `include "./voting.v"
3 `include "./voting_syn.v"
4 `include "./N16ADFP_StdCell.v"
```

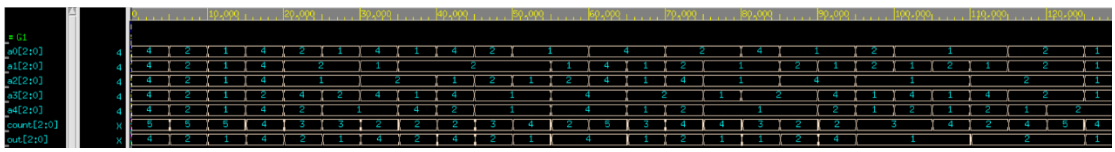
Inside the module, insert **initial $sdf_annotate ("voting_syn.sdf", u_voting);** to apply timing information from the SDF file to the design under test, ensuring that the synthesized design and standard cell library are correctly included for post-synthesis simulation.

```
1 voting u_voting (a0, a1, a2, a3, a4, out, count);
2 initial $sdf_annotate ("voting_syn.sdf", u_voting);
```

21. In the Xshell terminal, enter **ncverilog tb_voting.v +access+r** to rerun the simulation.



22. In Notepad++, check that the message **“Annotation completed successfully”** is displayed.
23. In the Xshell terminal, enter **nWave &** to launch nWave, open tb, and select the signals you want to observe.



III. Median Circuit

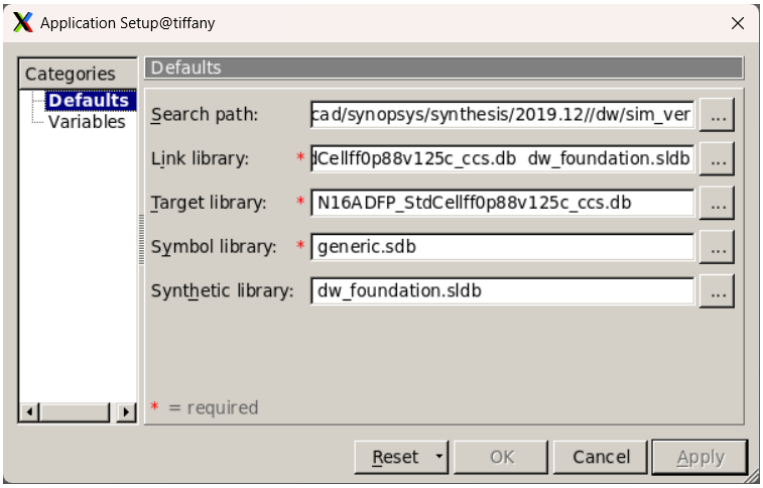
1. Upload the **.synopsys_dc.setup** file to the RCA/ directory via FileZilla.
2. In the Xshell terminal, enter **ls -al** to verify that the correct configuration file is included.

```
[B123015021@tiffany median]$ ls -a
.  ..  median.v  N16ADFP_StdCell.v  .synopsys_dc.setup  tb_median.v
```

3. In the Xshell terminal, enter **ncverilog tb_median.v +access+r** to compile and simulate top.v, and verify that the simulation runs correctly.



4. In the Xshell terminal, enter **dv &** and wait for Design Vision to launch.
5. In the Toolbox, select **File → Setup** and verify that the library contents are correctly configured.



6. Load the top.v file to be synthesized, and verify that the message **“Presto compilation completed successfully”** is displayed.

```
Presto compilation completed successfully.
Current design is now '/home/UGP/B123015021/HW3/HW2/median/onehot_to_bin.db:onehot_to_bin'
Loaded 4 designs.
Current design is 'onehot_to_bin'.
```

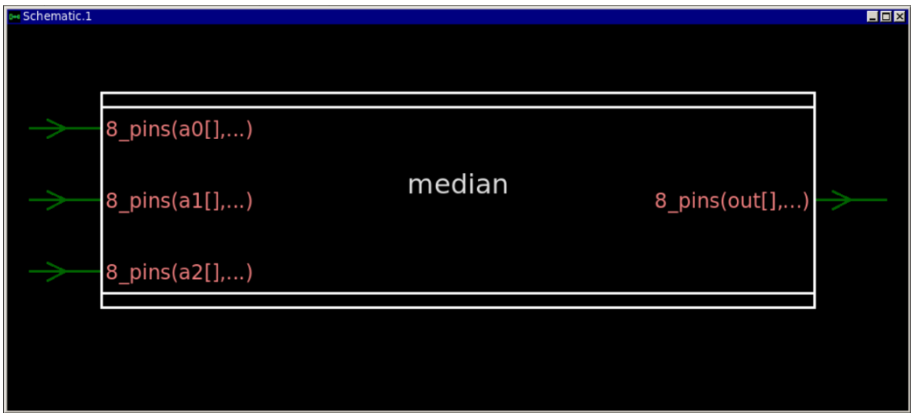
7. Verify that the hierarchy is set to the top module. Otherwise, enter the command **current_design median** in the command line.

```
design_vision> current_design median
Current design is 'median'.
```

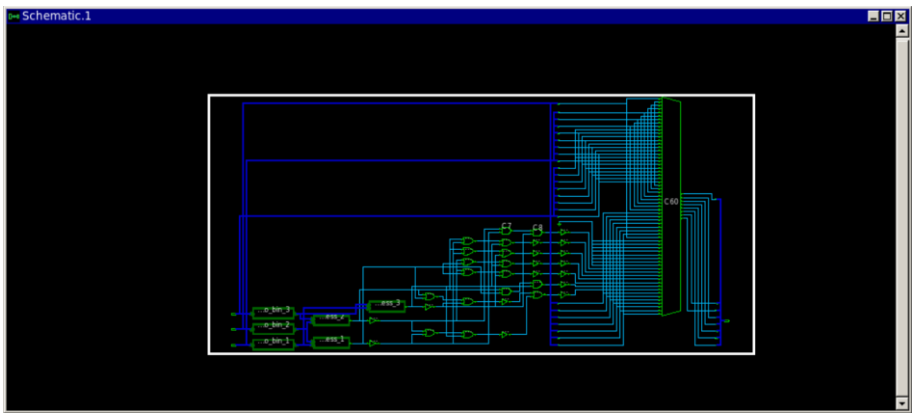
8. In the command line, enter: **set_fix_multiple_port_nets -all -buffer_constants**

```
design_vision> set_fix_multiple_port_nets -all -buffer_constants
1
```

9. View the Symbol View.



10. View the Schematic View.



11. In the Toolbox, select **Design → Compile Design**, and verify that the message **"Optimization Complete"** is displayed.

```
Optimization Complete
---------------------
1
```

12. In the Toolbox, select **Design → Report Area** to view the area report.

```
****************************************
Report : area
Design : median
Version: Q-2019.12
Date   : Sat Nov 15 02:16:41 2025
****************************************

Information: Updating design information... (UID-85)
Library(s) Used:

    N16ADFP_StdCellff0p88v125c_ccs (File: /cad/CBDK/Executable_Packa

Number of ports:                          86
Number of nets:                          129
Number of cells:                          60
Number of combinational cells:            54
Number of sequential cells:                0
Number of macros/black boxes:              0
Number of buf/inv:                         8
Number of references:                     12

Combinational area:               16.692480
Buf/Inv area:                      1.244160
Noncombinational area:             0.000000
Macro/Black Box area:              0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                  16.692480
Total area:                 undefined

***** End Of Report *****
```

13. In the Toolbox, select **Design → Report Power** to view the power report.

```
  Cell Internal Power  =   4.9899 uW   (61%)
  Net Switching Power  =   3.1435 uW   (39%)
                         ---------
Total Dynamic Power    =   8.1334 uW  (100%)

Cell Leakage Power     =   4.2324 uW

Information: report_power power group summary does not include estimated clock tree power. (PWR-789)

                 Internal         Switching           Leakage            Total
Power Group      Power            Power               Power              Power   (   %    )  Attrs
--------------------------------------------------------------------------------------------------
io_pad             0.0000           0.0000              0.0000             0.0000  (   0.00%)
memory             0.0000           0.0000              0.0000             0.0000  (   0.00%)
black_box          0.0000           0.0000              0.0000             0.0000  (   0.00%)
clock_network      0.0000           0.0000              0.0000             0.0000  (   0.00%)
register           0.0000           0.0000              0.0000             0.0000  (   0.00%)
sequential         0.0000           0.0000              0.0000             0.0000  (   0.00%)
combinational  4.9899e-03        3.1435e-03          4.2324e+03         1.2366e-02  ( 100.00%)
--------------------------------------------------------------------------------------------------
Total          4.9899e-03 mW     3.1435e-03 mW     4.2324e+03 nW     1.2366e-02 mW

***** End Of Report *****
```

14. In the Toolbox, select **Timing → Report Timing Path** to view the delay time from the path's starting point to each node.

    i. max

```
Startpoint: a0[3] (input port)
Endpoint: out[7] (output port)
Path Group: (none)
Path Type: max

Des/Clust/Port     Wire Load Model       Library
------------------------------------------------
onehot_to_bin_0    ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
median             ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
cmp3_less_2        ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
cmp3_less_1        ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs

Point                                                   Incr       Path
--------------------------------------------------------------------------
input external delay                                    0.00       0.00 f
a0[3] (in)                                              0.00       0.00 f
onehot_to_bin_1/in[3] (onehot_to_bin_0)                 0.00       0.00 f
onehot_to_bin_1/U3/Z (OR4D1BWP16P90)                    0.02       0.02 f
onehot_to_bin_1/out[0] (onehot_to_bin_0)                0.00       0.02 f
cmp3_less_1/B[0] (cmp3_less_2)                          0.00       0.02 f
cmp3_less_1/U5/ZN (INR2D1BWP16P90)                      0.01       0.04 f
cmp3_less_1/U3/Z (AN2D1BWP16P90)                        0.01       0.05 f
cmp3_less_1/U2/ZN (OAI222D1BWP16P90)                    0.01       0.06 r
cmp3_less_1/U1/ZN (IOA21D1BWP16P90)                     0.02       0.07 f
cmp3_less_1/L (cmp3_less_2)                             0.00       0.07 f
U51/ZN (CKND1BWP16P90)                                  0.01       0.08 r
U45/ZN (ND2D1BWP16P90)                                  0.01       0.09 f


U43/ZN (MUX2ND1BWP16P90)                                0.03       0.12 r
U28/ZN (AOI22D1BWP16P90)                                0.01       0.13 f
U27/ZN (IOA21D1BWP16P90)                                0.00       0.14 r
out[7] (out)                                            0.00       0.14 r
data arrival time                                                  0.14
--------------------------------------------------------------------------
(Path is unconstrained)
```

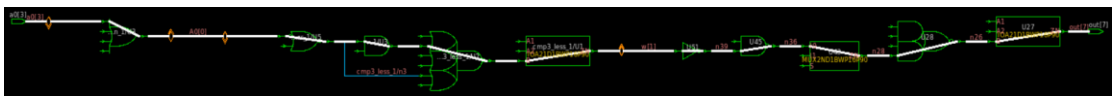
ii. min

```
Startpoint: a2[7] (input port)
Endpoint: out[7] (output port)
Path Group: (none)
Path Type: min

Des/Clust/Port     Wire Load Model       Library
------------------------------------------------
onehot_to_bin_1    ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
median             ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs

Point                                   Incr       Path
------------------------------------------------------------
input external delay                    0.00       0.00 r
a2[7] (in)                              0.00       0.00 r
U28/ZN (AOI22D1BWP16P90)                0.00       0.00 f
U27/ZN (IOA21D1BWP16P90)                0.00       0.01 r
out[7] (out)                            0.00       0.01 r
data arrival time                                  0.01
------------------------------------------------------------
(Path is unconstrained)
```

15. In the Toolbox, select **Select → Path From/Through/To**, set the delay type to max, and check the critical path.



16. In the command line, enter: **change_names -hierarchy -rule verilog**

```
design_vision> change_names -hierarchy -rule verilog
1
```

17. In the command line, enter: **report_qor > median.qor** to save the report.

```
design_vision> report_qor > median.qor
```

18. In the Toolbox, select **File → Save As**, name the file **median_syn.v**, and click **Save** to store the synthesized file.

19. In the command line, enter: **write_sdf -version 1.0 -context verilog median_syn.sdf** to write the timing information into the SDF file.

```
design_vision> write_sdf -version 1.0 -context verilog median_syn.sdf
Information: Annotated 'cell' delays are assumed to include load delay. (UID-282)
Information: Writing timing information to file '/home/UGP/B123015021/HW3/HW2/median/median_syn.sdf'. (WT-3)
1
```

20. Modify the testbench by commenting out **`include "./median.v"**, then adding **`include "./median_syn.v"** and **`include "./N16ADFP_StdCell.v"**.

```
1 `timescale 10ns/100ps
2 // `include "./median.v"
3 `include "./median_syn.v"
4 `include "./N16ADFP_StdCell.v"
```

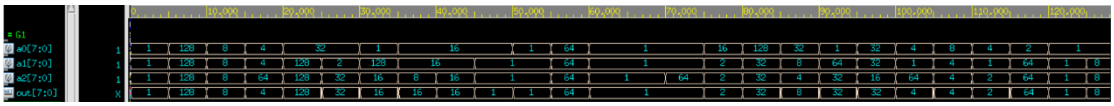
Inside the module, insert **initial $sdf_annotate ("median_syn.sdf", u_median);** to apply timing information from the SDF file to the design under test, ensuring that the synthesized design and standard cell library are correctly included for post-synthesis simulation.

```
1 median u_median (a0, a1, a2, out);
2 initial $sdf_annotate ("median_syn.sdf", u_median);
```

21. In the Xshell terminal, enter **ncverilog tb_median.v +access+r** to rerun the simulation.



22. In Notepad++, check that the message **"Annotation completed successfully"** is displayed.
23. In the Xshell terminal, enter **nWave &** to launch nWave, open tb, and select the signals you want to observe.



➢ Circuit Analysis

I. Voting Circuit

1. Area Report

```
****************************************
Report : area
Design : voting
Version: Q-2019.12
Date   : Sat Nov 15 01:55:28 2025
****************************************

Information: Updating design information... (UID-85)
Library(s) Used:

    N16ADFP_StdCellff0p88v125c_ccs (File: /cad/CBDK/Executable_Packa

Number of ports:                           66
Number of nets:                           120
Number of cells:                           66
Number of combinational cells:             60
Number of sequential cells:                 0
Number of macros/black boxes:               0
Number of buf/inv:                         12
Number of references:                      11

Combinational area:                18.766081
Buf/Inv area:                       1.866240
Noncombinational area:              0.000000
Macro/Black Box area:               0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                   18.766081
Total area:                 undefined

***** End Of Report *****
```

In the ADFP process, the area of a single NAND gate is around 0.20736 μm². Therefore, to estimate the number of NAND gates, we can divide the total circuit area by this number:

$$Number\ of\ NAND\ Gates = \frac{18.766081}{0.20736} \approx 90.5$$

2. Power Report

```
  Cell Internal Power  =  12.0654 uW   (56%)
  Net Switching Power  =   9.3089 uW   (44%)
                         ---------
Total Dynamic Power    =  21.3743 uW  (100%)

Cell Leakage Power     =   5.8527 uW

Information: report_power power group summary does not include estimated clock tree power. (PWR-789)

                 Internal         Switching           Leakage            Total
Power Group      Power            Power               Power              Power   (   %    )  Attrs
--------------------------------------------------------------------------------------------------
io_pad             0.0000           0.0000            0.0000             0.0000  (   0.00%)
memory             0.0000           0.0000            0.0000             0.0000  (   0.00%)
black_box          0.0000           0.0000            0.0000             0.0000  (   0.00%)
clock_network      0.0000           0.0000            0.0000             0.0000  (   0.00%)
register           0.0000           0.0000            0.0000             0.0000  (   0.00%)
sequential         0.0000           0.0000            0.0000             0.0000  (   0.00%)
combinational  1.2065e-02       9.3089e-03        5.8527e+03         2.7227e-02  ( 100.00%)
--------------------------------------------------------------------------------------------------
Total          1.2065e-02 mW     9.3089e-03 mW     5.8527e+03 nW     2.7227e-02 mW
```

$$Power\ Consumption = 21.3743\ \mu W$$

3. Timing Report

```
Startpoint: a2[2] (input port)
Endpoint: count[0] (output port)
Path Group: (none)
Path Type: max

Des/Clust/Port     Wire Load Model       Library
------------------------------------------------
voting             ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
add5_1             ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
cmp3_less_1        ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
cmp3_less_0        ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs

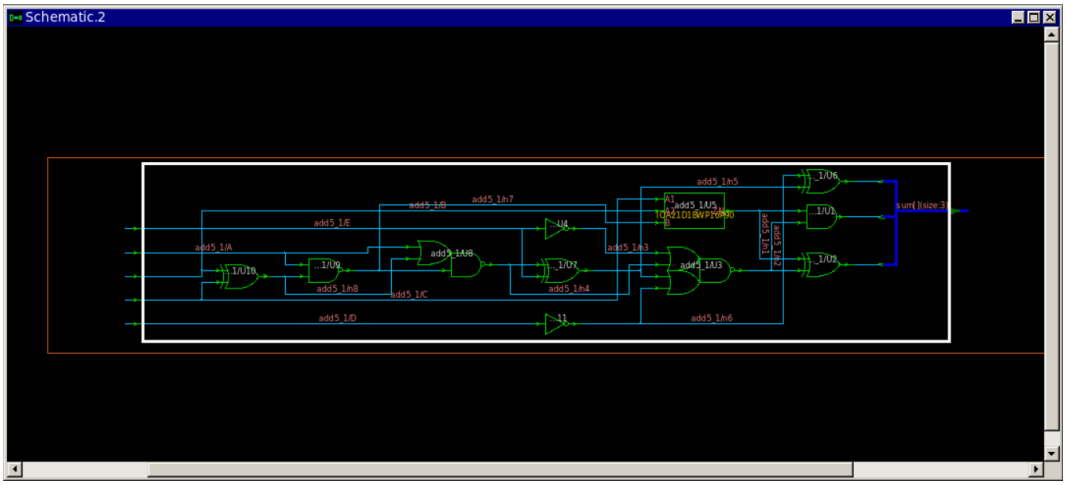
Point                                                   Incr       Path
--------------------------------------------------------------------------
input external delay                                    0.00       0.00 f
a2[2] (in)                                              0.00       0.00 f
add5_3/C (add5_1)                                       0.00       0.00 f
add5_3/U10/Z (XOR2D1BWP16P90)                           0.02       0.02 r
add5_3/U9/ZN (ND2D1BWP16P90)                            0.01       0.03 f
add5_3/U8/ZN (OAI21D1BWP16P90)                          0.01       0.04 r
add5_3/U7/Z (XOR2D1BWP16P90)                            0.02       0.06 f
add5_3/U3/ZN (OAI22D1BWP16P90)                          0.01       0.07 r
add5_3/U2/Z (XOR2D1BWP16P90)                            0.03       0.10 f
add5_3/sum[1] (add5_1)                                  0.00       0.10 f
cmp3_less_1/A[1] (cmp3_less_0)                          0.00       0.10 f
cmp3_less_1/U4/ZN (CKND1BWP16P90)                       0.01       0.11 r
cmp3_less_1/U3/Z (AN2D1BWP16P90)                        0.01       0.12 r

cmp3_less_1/U2/ZN (OAI222D1BWP16P90)                    0.01       0.13 f
cmp3_less_1/U1/ZN (IOA21D1BWP16P90)                     0.01       0.14 r
cmp3_less_1/L (cmp3_less_0)                             0.00       0.14 r
U17/ZN (INR2D1BWP16P90)                                 0.02       0.16 r
U16/ZN (AOI22D1BWP16P90)                                0.01       0.17 f
U15/ZN (IOA21D1BWP16P90)                                0.00       0.17 r
count[0] (out)                                          0.00       0.17 r
data arrival time                                                  0.17
--------------------------------------------------------------------------
(Path is unconstrained)
```

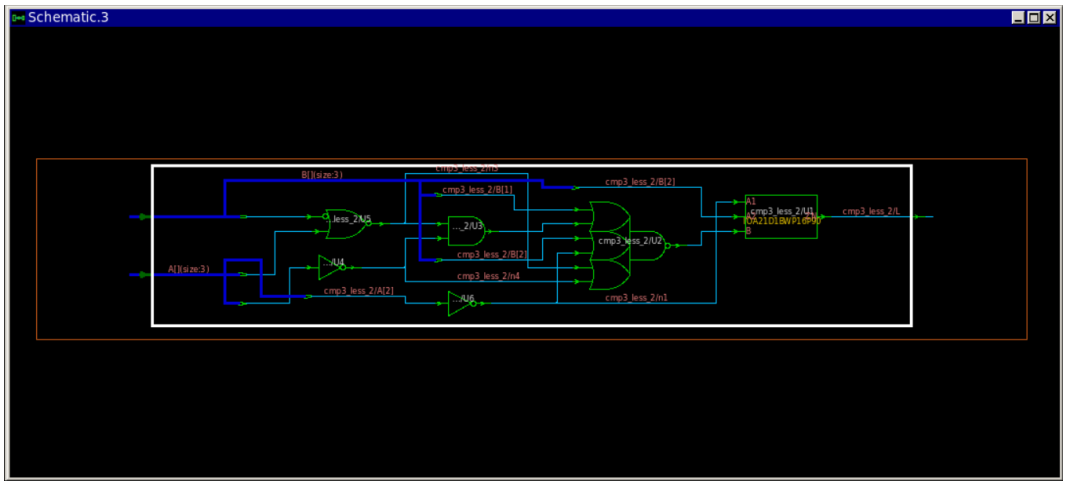
The voting circuit has a total of 19 logic levels, with a maximum time delay of 0.17 ns.

4. Schematic View

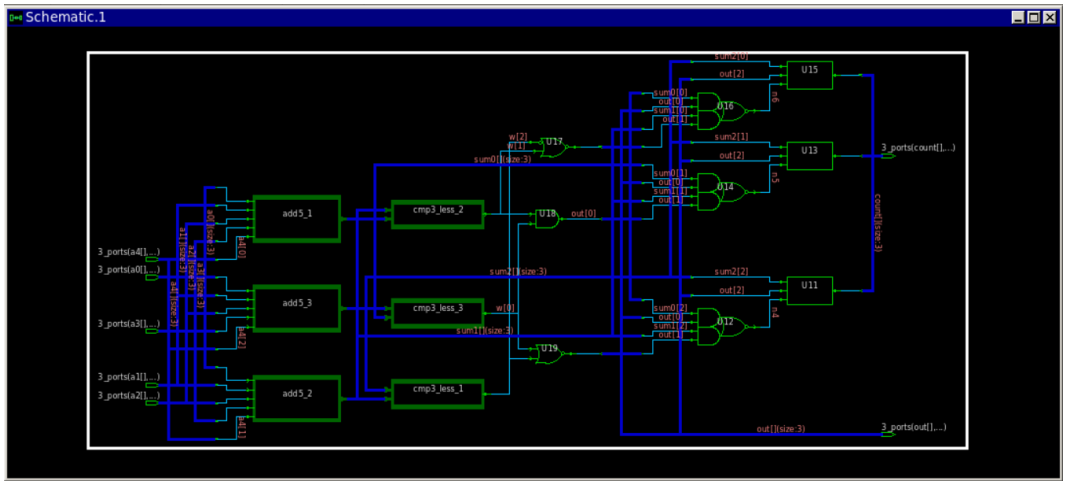
i. Adder



ii. Comparator



iii. Voting Circuit

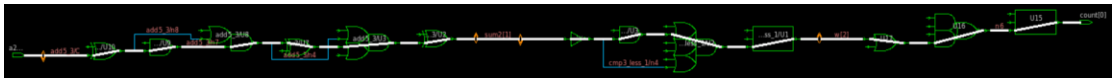


Each adder contains 16 gates and each comparator contains 11 gates.

With 18 additional gates at the top level, the total becomes:

$$3 \times 16 + 3 \times 11 + 18 = 99\ Gates$$

5. Critical Path



$$Critical\ Path = 44\Delta$$

II. Median Circuit

1. Area Report

```
****************************************
Report : area
Design : median
Version: Q-2019.12
Date   : Sat Nov 15 02:16:41 2025
****************************************

Information: Updating design information... (UID-85)
Library(s) Used:

    N16ADFP_StdCellff0p88v125c_ccs (File: /cad/CBDK/Executable_Packa

Number of ports:                           86
Number of nets:                           129
Number of cells:                           60
Number of combinational cells:             54
Number of sequential cells:                 0
Number of macros/black boxes:               0
Number of buf/inv:                          8
Number of references:                      12

Combinational area:                16.692480
Buf/Inv area:                       1.244160
Noncombinational area:              0.000000
Macro/Black Box area:               0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                   16.692480
Total area:                 undefined

***** End Of Report *****
```

In the ADFP process, the area of a single NAND gate is around 0.20736 μm². Therefore, to estimate the number of NAND gates, we can divide the total circuit area by this number:

$$Number\ of\ NAND\ Gates = \frac{16.692480}{0.20736} \approx 80.5$$

2. Power Report

```
  Cell Internal Power  =   4.9899 uW   (61%)
  Net Switching Power  =   3.1435 uW   (39%)
                         ---------
Total Dynamic Power    =   8.1334 uW  (100%)

Cell Leakage Power     =   4.2324 uW

Information: report_power power group summary does not include estimated clock tree power. (PWR-789)

                 Internal         Switching           Leakage            Total
Power Group      Power            Power               Power              Power   (   %    )  Attrs
--------------------------------------------------------------------------------------------------
io_pad             0.0000           0.0000            0.0000             0.0000  (   0.00%)
memory             0.0000           0.0000            0.0000             0.0000  (   0.00%)
black_box          0.0000           0.0000            0.0000             0.0000  (   0.00%)
clock_network      0.0000           0.0000            0.0000             0.0000  (   0.00%)
register           0.0000           0.0000            0.0000             0.0000  (   0.00%)
sequential         0.0000           0.0000            0.0000             0.0000  (   0.00%)
combinational  4.9899e-03        3.1435e-03        4.2324e+03         1.2366e-02  ( 100.00%)
--------------------------------------------------------------------------------------------------
Total          4.9899e-03 mW     3.1435e-03 mW     4.2324e+03 nW      1.2366e-02 mW

***** End Of Report *****
```

$$Power\ Consumption = 8.1334\ \mu W$$

3. Timing Report

```
Startpoint: a0[3] (input port)
Endpoint: out[7] (output port)
Path Group: (none)
Path Type: max

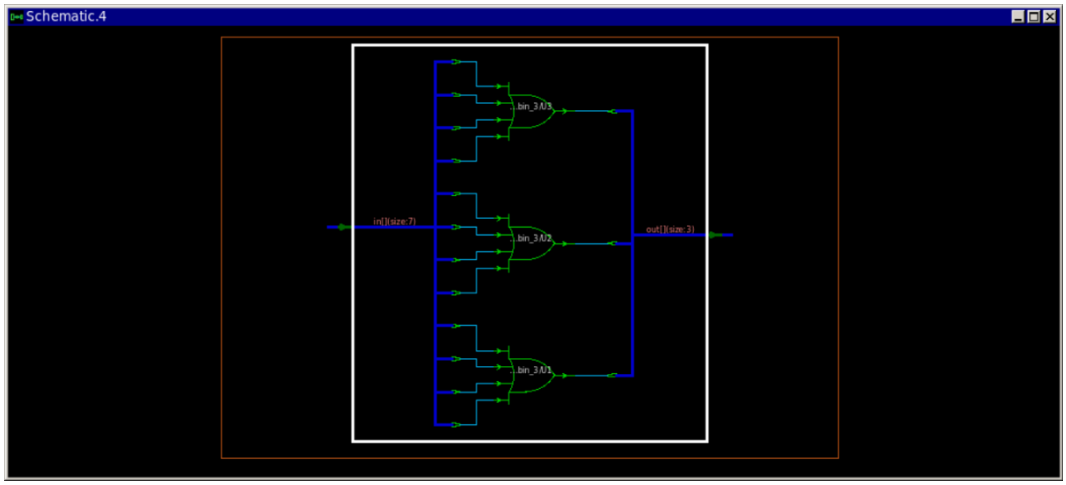
Des/Clust/Port     Wire Load Model       Library
------------------------------------------------
onehot_to_bin_0    ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
median             ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
cmp3_less_2        ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs
cmp3_less_1        ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs

Point                                                   Incr       Path
--------------------------------------------------------------------------
input external delay                                    0.00       0.00 f
a0[3] (in)                                              0.00       0.00 f
onehot_to_bin_1/in[3] (onehot_to_bin_0)                 0.00       0.00 f
onehot_to_bin_1/U3/Z (OR4D1BWP16P90)                    0.02       0.02 f
onehot_to_bin_1/out[0] (onehot_to_bin_0)                0.00       0.02 f
cmp3_less_1/B[0] (cmp3_less_2)                          0.00       0.02 f
cmp3_less_1/U5/ZN (INR2D1BWP16P90)                      0.01       0.04 f
cmp3_less_1/U3/Z (AN2D1BWP16P90)                        0.01       0.05 f
cmp3_less_1/U2/ZN (OAI222D1BWP16P90)                    0.01       0.06 r
cmp3_less_1/U1/ZN (IOA21D1BWP16P90)                     0.02       0.07 f
cmp3_less_1/L (cmp3_less_2)                             0.00       0.07 f
U51/ZN (CKND1BWP16P90)                                  0.01       0.08 r
U45/ZN (ND2D1BWP16P90)                                  0.01       0.09 f
U43/ZN (MUX2ND1BWP16P90)                                0.03       0.12 r
U28/ZN (AOI22D1BWP16P90)                                0.01       0.13 f
U27/ZN (IOA21D1BWP16P90)                                0.00       0.14 r
out[7] (out)                                            0.00       0.14 r
data arrival time                                                  0.14
--------------------------------------------------------------------------
(Path is unconstrained)
```

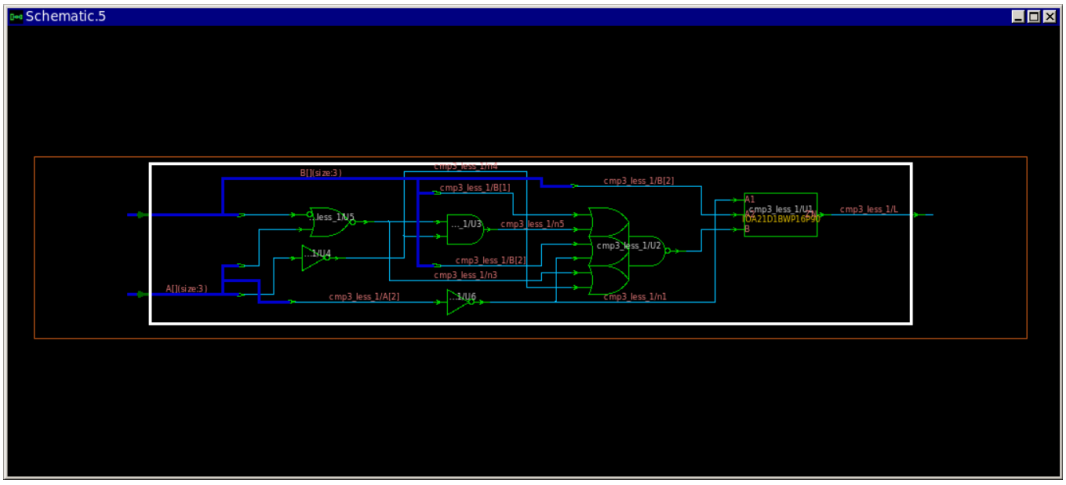
The median circuit has a total of 16 logic levels, with a maximum time delay of 0.14 ns.

4. Schematic View

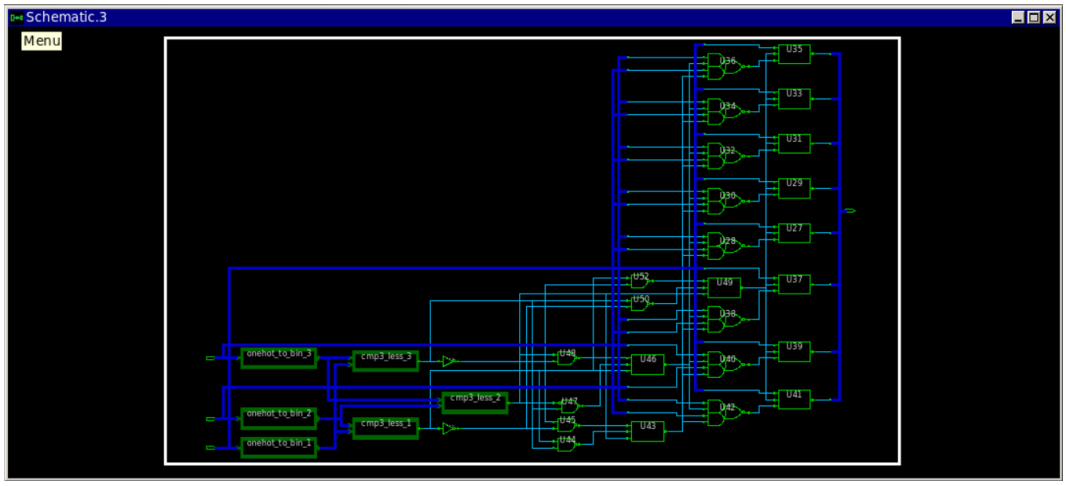
i. Encoder



ii. Comparator



iii. Median Circuit

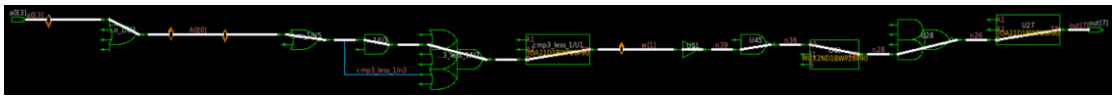


Each encoder contains 3 gates and each comparator contains 11 gates.

With 57 additional gates at the top level, the total becomes:

$$3 \times 3 + 3 \times 10 + 57 = 96\ Gates$$

5. Critical Path



$$Critical\ Path = 29\Delta$$

III. Comparison

1. Voting Circuit

	Pre-synthesized	Post-synthesized
Number of Gates	90	99
Logic Levels	13	16
Critical Path	41Δ	44Δ

2. Median Circuit

	Pre-synthesized	Post-synthesized
Number of Gates	89	96
Logic Levels	14	19
Critical Path	21Δ	29Δ

3. Conclusion

The post-synthesized circuit has more gates, deeper logic levels, and a longer critical path, likely because it must be built using the standard-cell library, whose components contain more internal Boolean logic than the simplified gates used in the pre-synthesis model.

Although our analytical estimates appear worse than the pre-synthesized results, practical post-synthesis implementations typically perform better, since real

standard-cell components are highly optimized and typically do not follow the gate-count and delay assumptions used in our analysis.

➢ Testbench Design

I. Adder

The testbench instantiates the DUT with two 14-bit inputs A, B, a 1-bit mode input, a 14-bit sum output, and a 1-bit carry-out. A 15-bit internal variable expect is used as a golden reference: when mode = 1, the testbench computes expect = A + B; when mode = 0, it computes two's-complement subtraction expect = A + (~B) + 1. The testbench then compares {c_out, sum} against expect and records pass/fail counts. The run_directed task applies a set of chosen corner cases such as 0+0, long carry chain, overflow, while run_random generates a large number of random operations.

II. Voting Circuit

The DUT receives five 3-bit one-hot inputs a0~a4, and produces a 3-bit one-hot winner out together with a 3-bit output count. A golden reference counts how many votes each one-hot position receives by summing the corresponding bits of i0–i4, then selects the candidate with the largest count and outputs both the expected one-hot result and its vote count. The check task then drives the DUT inputs, calls the golden model, and compares {out, count} with the expected values, updating pass/fail counters and printing detailed failing messages on any mismatch. The run_directed task covers key patterns such as 5-0-0, 4-1-0, 3-1-1 distributions, and tie cases like 2–2–1, while run_random applies a large number of randomized one-hot combinations.

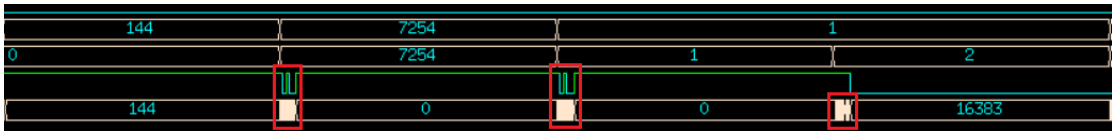
III. Median Circuit

The DUT takes three 8-bit one-hot inputs a0~a2 and outputs an 8-bit one-hot out corresponding to the median among the three inputs. The testbench defines a function oh(idx) to generate one-hot vectors for indices 0–7, and an idx8 function to decode a 1-hot vector back to its index. The golden model golden first converts each input

vector to its integer index x, y, z, computes the minimum, maximum, then calculate the median index md = x + y + z − min − max, and finally converting md back to a one-hot vector via oh(md). The check task compares the DUT output with the golden result, updating pass/fail counters and printing detailed error messages on mismatch. Directed tests in run_directed cover corner cases like all-equal, two-equal, while run_random generates a series of random one-hot inputs.

➢ Delay Effect

When observing the waveform of the circuits, I noticed that after synthesis, the circuit sometimes produced tiny spikes or brief unknown values right when the inputs changed.



This is probably because that signals don't all settle at the same time once real delays are introduced in practical circuits. Therefore, when designing circuits, we should avoid creating unnecessarily deep logic paths, because excessive delay and uneven signal arrival times can introduce glitches or timing hazards that may compromise the functionality of the design, even if the logic is correct.

To avoid accidentally treating these harmless glitches as errors during comparison with the golden model, I added a #5 wait before checking the outputs in each cycle. This gives the circuit time to settle, ensuring that the verification is based only on the final correct result.

➢ Discussion & Reflection

Through this homework, I found that designing circuits at the gate level is not always the best approach, especially when targeting a real standard-cell library. My initial implementation of the adder and comparator was written entirely in gate-level structure, and the synthesis tool could not effectively optimize this structure, resulting

in larger area and worse overall performance than expected. After rewriting the same functionality in a higher-level RTL style, the compiler was able to apply optimizations, which significantly reduce the total area while maintaining approximately the same maximum timing delay. This taught me that even when two designs represent the same hardware structure, coding style can greatly influence the final performance of the synthesized circuit.

This homework also helped me understand how a complete circuit design flow works from start to finish. From the Verilog design in HW2, to synthesizing it to gate-level hardware, applying SDF timing, and verifying the final design with directed and random testbenches. Seeing how the circuit evolved at each step made the whole process much more concrete and showed me how coding style, hardware structure, and synthesis decisions all interact. Overall, I gained a clearer view of how a full digital circuit design flow works and how proper design methodology directly affects the final quality of a hardware implementation.